

EOB Extractor — Technical Handoff

Project: eob-extractor

Date: 2026-06-08

Status: Production-ready (audit findings tracked below)

Owner: engineering

Compliance: HIPAA

1. Overview

EOB Extractor is a serverless pipeline that automatically extracts insurance company and arbitration contact information from Explanation of Benefits (EOB) PDF documents. It ingests PDFs uploaded to S3, classifies them using Amazon Bedrock, extracts structured data, validates it against a Zod schema and business rules, cross-references an existing insurance contact database, and stores results in DynamoDB. Low-confidence extractions are routed to a human review queue.

Purpose: Remove manual data-entry for arbitration contact discovery across EOB documents uploaded via the ClickUp integration.

2. Repository Structure

```
eob-extractor/  
├── src/  
│   ├── domain/  
│   │   ├── entities/eob-extraction.ts           # EobExtraction aggregate root  
│   │   ├── repositories/eob-extraction.repository.ts  
│   │   └── value-objects/  
│   │       ├── confidence-score.ts             # HIGH>=0.85, MEDIUM>=0.50, LOW<0.50  
│   │       └── extraction-status.ts             # EXTRACTED | REVIEW_PENDING | FAILED  
│   ├── application/  
│   │   ├── prompts/  
│   │   │   ├── classify-prompt.ts               # EOB_CLASSIFICATION_SYSTEM_PROMPT  
│   │   │   ├── extract-prompt.ts               # buildExtractionUserPrompt()  
│   │   │   └── system-prompt.ts  
│   │   ├── schemas/eob-extraction.schema.ts    # Zod v4 output schema  
│   │   └── validation/business-rules.ts  
│   ├── infrastructure/  
│   │   ├── bedrock/  
│   │   │   ├── bedrock-client.ts               # invokeModel(), InvokeModelCommand  
│   │   │   └── model-fallback.ts               # CLASSIFY_CHAIN, EXTRACT_CHAIN  
│   │   ├── logging/audit-logger.ts             # PHI_FIELDS blocklist, sanitizeS3Key()  
│   │   ├── persistence/dynamodb-eob.repository.ts  
│   │   └── storage/s3-pdf-reader.ts  
│   ├── handlers/  
│   │   ├── trigger.handler.ts                  # SQS → SFN start  
│   │   ├── validate-pdf.handler.ts  
│   │   ├── classify-eob.handler.ts  
│   │   ├── extract-eob.handler.ts  
│   │   ├── validate-data.handler.ts  
│   │   ├── lookup-insurance.handler.ts  
│   │   └── store-result.handler.ts  
│   └── lib/  
│       ├── eob-extractor-stack.ts              # CDK root stack  
│       └── constructs/  
│           ├── storage.construct.ts             # KMS keys, DynamoDB, S3  
│           ├── queuing.construct.ts             # SQS ingest/review/DLQ  
│           ├── monitoring.construct.ts          # SNS topics, CloudWatch alarms  
│           ├── extraction.construct.ts          # Lambda wiring + event sources  
│           ├── pipeline-functions.ts            # 6 pipeline Lambda definitions  
│           ├── extraction-state-machine.ts      # Step Functions definition  
│           └── dashboard.construct.ts  
└── test/  
    └── unit/                                    # Vitest unit tests per handler
```

3. Tech Stack

Layer	Technology	Version
Runtime	Node.js	20.x
Language	TypeScript	~5.9.3
IaC	AWS CDK	2.1112.0
Schema validation	Zod	v4 (^4.3.6)

Layer	Technology	Version
ID generation	ULID	^3.0.2
Test framework	Vitest	^4.1.1
Bundler	esbuild	^0.27.4

4. AWS Resources Summary

Service	Resource	Purpose
Lambda (7)	eob-trigger	SQS consumer; starts SFN execution
Lambda	eob-validate-pdf	Validates PDF existence/format
Lambda	eob-classify-eob	Bedrock classify chain
Lambda	eob-extract-eob	Bedrock extract chain
Lambda	eob-validate-data	Zod + business-rule validation
Lambda	eob-lookup-insurance	Cross-ref Insurance-Arbitration-contacts
Lambda	eob-store-result	DynamoDB write + review queue
Step Functions	eob-extraction-pipeline	STANDARD; 15-min timeout; X-Ray; KMS logs
DynamoDB	eob-extractions	PAY_PER_REQUEST; KMS CMK; PITR; Streams
DynamoDB (ext)	Insurance-Arbitration-contacts	Pre-existing; read + conditional write
S3 (ext)	bucket-specialops / bucket-specialops-sandbox	PDF source; not CDK-managed
S3	eob-extractor-audit-{account}	Audit logs; Object Lock COMPLIANCE; 6yr
SQS	eob-extractor-ingest	S3 events → trigger; DLQ after 3 retries
SQS	eob-extractor-review	Low-confidence review (⚠ no DLQ — TD-005)
SQS	eob-extractor-dlq	Dead letters; 14-day retention
SNS	eob-extractor-ops-alerts	Operational alerts
SNS	eob-extractor-review-alerts	New/mismatch insurance contact alerts
KMS	alias/eob-extractor/phi	DynamoDB + Lambda env encryption
KMS	alias/eob-extractor/audit	S3 audit + SFN log group encryption
Bedrock	us.anthropic.claude-haiku-4-5	Classify primary
Bedrock	us.anthropic.claude-sonnet-4-20250514	Classify fallback; Extract fallback 2
Bedrock	us.anthropic.claude-sonnet-4-6	Extract primary

5. Environment Variables (per Lambda)

Lambda	Env Var	Value
eob-trigger	STATE_MACHINE_ARN	eob-extraction-pipeline ARN
eob-trigger	TABLE_NAME	eob-extractions
eob-validate-pdf	BUCKET_NAME	bucket-specialops[-sandbox]
eob-classify-eob	BUCKET_NAME	bucket-specialops[-sandbox]
eob-extract-eob	BUCKET_NAME	bucket-specialops[-sandbox]
eob-lookup-insurance	CONTACTS_TABLE_NAME	Insurance-Arbitration-contacts
eob-lookup-insurance	NOTIFY_TOPIC_ARN	eob-extractor-review-alerts ARN
eob-store-result	TABLE_NAME	eob-extractions
eob-store-result	REVIEW_QUEUE_URL	eob-extractor-review URL

6. S3 Trigger Path

The pipeline activates on S3 `OBJECT_CREATED` events matching: - **Bucket:** `bucket-specialops` (prod) or `bucket-specialops-sandbox` (dev) - **Prefix:** `clickup/` - **Suffix:** `.pdf`

Documents outside this path are ignored by the event notification filter.

7. Confidence Score Thresholds

Score	Level	Status	Routing
≥ 0.85	HIGH	EXTRACTED	LookupInsurance → StoreExtracted
0.50 – 0.84	MEDIUM	REVIEW_PENDING	LookupInsurance → StoreReviewPending
< 0.50	LOW	FAILED	StoreFailed (no lookup)

8. Open Audit Findings

ID	Severity	Title	Status
#001	HIGH	S3 PatientName in keys	FIXED (audit/2026-06-04 — <code>sanitizeS3Key()</code>)
#002	HIGH	Lambda functions NOT VPC-attached	OPEN — architectural gap
#004	MEDIUM	PHI blocklist incomplete in sanitizers	OPEN
#005	MEDIUM	eob-extractor-review queue has no DLQ	OPEN (TD-005)

ID	Severity	Title	Status
#006	MEDIUM	Bedrock inference not logged with user identity	OPEN
#007	MEDIUM	Insurance-Arbitration-contacts read without PITR/audit trail	OPEN
#008	LOW	ULID generation not seeded from KMS	OPEN
#009	LOW	No IaC for bucket-specialops (pre-existing, manual)	OPEN

Critical path for next sprint: #002 (VPC), #004 (PHI blocklist), #005 (review DLQ).

9. CDK Deployment

```
# Install dependencies
npm ci

# Synthesize (review before deploying)
npx cdk synth --context environment=sandbox --context bucketName=bucket-specialops-sandbox

# Deploy sandbox
npx cdk deploy --context environment=sandbox --context bucketName=bucket-specialops-sandbox

# Deploy production
npx cdk deploy --context environment=production --context bucketName=bucket-specialops
```

CDK context keys: - `environment` — `sandbox` | `production` (controls RemovalPolicy + deletionProtection) - `bucketName` — source PDF bucket name - `contactsTable` — external contacts table (default: `Insurance-Arbitration-contacts`) - `costCenter` — cost allocation tag (default: `engineering`)

10. Running Tests

```
npm run test          # all tests via Vitest
npm run test:unit     # unit tests only (test/unit/)
npm run test:watch    # watch mode
```

11. Architecture Decision Records

Decision	Rationale
Step Functions STANDARD (not EXPRESS)	Execution history retention for audit; 15-min timeout sufficient
Bedrock fallback chain	On-demand quotas per model; chain ensures completion under throttling

Decision	Rationale
ULID correlation IDs	Time-sortable, collision-resistant, no external dependency
Zod v4 schema validation	Runtime type safety at Lambda boundary; schema-as-truth for extraction contract
<code>includeExecutionData: false</code> on SFN logs	Prevents PHI (PDF content, extraction JSON) from appearing in CloudWatch
<code>attribute_not_exists</code> condition on DynamoDB PutItem	Idempotent writes; concurrent executions cannot overwrite existing extraction records
DLQ ingest <code>maxReceiveCount=3</code>	After 3 visibility-timeout expiries the message routes to DLQ for inspection